

# 2025 UT Austin Invitational Science Olympiad

## Forensics C Exam Packet

October 25th, 2025



Student Name(s): \_\_\_\_\_

Team Number & School Name: \_\_\_\_\_

### Parameters:

- Each participant may bring 1 sheet of paper and lab equipment specified in the Science Olympiad official rules.
- Each participant is required to wear proper PPE at ALL times.
- Failure to clean up your station before the conclusion of the 50 minutes will result in a penalty of up to 10%.

### Directions:

- You will have 50 minutes to complete this exam.
- All answers must be indicated on your answer sheet.
- Point values indicated for all questions are indicated in test booklets.
- The maximum number of obtainable points on this exam is 292.

**Test Writer:** Ai Vy Le, *UT Austin Neuroscience '24*

**Artist Credits:** Martin Nguyen, *UT Austin Computer Science '23*

Any questions after the conclusion of the event may be directed to [aivyvy@utexas.edu](mailto:aivyvy@utexas.edu)

## Qualitative Analysis

1. (70 points) Using the samples provided, fill out each powder's respective chemical formula, name, and one applicable connection between this powder and the story.

| Powder # | Chemical Formula (2)<br>Write NA if not applicable | Powder Name (2) | Application in Story (1) |
|----------|----------------------------------------------------|-----------------|--------------------------|
| 1        |                                                    |                 |                          |
| 2        |                                                    |                 |                          |
| 3        |                                                    |                 |                          |
| 4        |                                                    |                 |                          |
| 5        |                                                    |                 |                          |
| 6        |                                                    |                 |                          |
| 7        |                                                    |                 |                          |
| 8        |                                                    |                 |                          |
| 9        |                                                    |                 |                          |
| 10       |                                                    |                 |                          |
| 11       |                                                    |                 |                          |
| 12       |                                                    |                 |                          |
| 13       |                                                    |                 |                          |
| 14       |                                                    |                 |                          |

2. (2 points) Write the complete, balanced chemical reaction between powder 14 and sodium hydroxide. Include states of matter.
3. (1 point) Why is acetic acid ( $\text{CH}_3\text{COOH}$ ) considered a weak acid?
4. (1 point) Why does a solute dissolve faster in a solvent when it is ground into smaller pieces?

5. (1 point) Soap molecules \_\_\_\_\_ oils and dirt, allowing them to be removed from surfaces.
6. (1 point) What happens when a metal carbonate reacts with an acid?
7. (2 points) Luminol is commonly used in forensic science because it reacts with \_\_\_\_\_ in blood to produce a \_\_\_\_\_ during chemiluminescence.
8. (2 points) Explain why silver is considered the best conductor of electricity among metals.
9. (2 points) Fermentation is an anaerobic process that converts glucose into energy and other products. Identify one common byproduct of fermentation in muscle cells and one in yeast.
10. (2 points) The process of \_\_\_\_\_ involves the conversion of ammonia into nitrites and then nitrates by bacteria, making nitrogen available for plant uptake.

## Polymers

11. (24 points) Using the information found in your image sheet, fill out the table with each plastic's respective information. \*Hint\* Plastic 2 is more recyclable than Plastic 6.

| Plastic # | Abbreviated Name (1) | IUPAC ID & Recycling Code(1) | Formula (1) |
|-----------|----------------------|------------------------------|-------------|
| 1         |                      |                              |             |
| 2         |                      |                              |             |
| 3         |                      |                              |             |
| 4         |                      |                              |             |
| 5         |                      |                              |             |
| 6         |                      |                              |             |
| 7         |                      |                              |             |
| 8         |                      |                              |             |

12. (1 point) Polypropylene (PP) and Polyethylene (PE) are both thermoplastics. Give one reason why PP might be chosen over PE for certain applications.
13. (2 points) Explain why thermoset plastics are often used for electrical insulation, whereas thermoplastics are more common in packaging. Categorize plastics 1-8 based on their thermal properties.

14. (2 points) What do the numbers 1-7 inside the recycling symbol on plastic products represent? How does category 7 differ from categories 1-6?
15. (2 points) Polyethylene (PE) and Polystyrene (PS) produce different flame colours when burned. Identify each flame colour and explain the reason for the difference.
16. (1 point) In a flame test, plastics containing halogen atoms, such as chlorine or bromine, often produce a \_\_\_\_\_ and incomplete \_\_\_\_\_ due to the halogen interfering with the burning process.
17. (20 points) Using the information found in your image sheet, fill out each fiber's name, whether it is organic or synthetic, and one applicable connection between this fiber and the story.

| Fiber # | Fiber Name (2) | Organic or Synthetic (1) | Application in Story (2) |
|---------|----------------|--------------------------|--------------------------|
| 1       |                |                          |                          |
| 2       |                |                          |                          |
| 3       |                |                          |                          |
| 4       |                |                          |                          |

18. (1 point) If fibers 1–4 are organic, categorize them as either animal-based or plant-based.
19. (1 point) What is the key difference between animal-based and plant-based fibers?
20. (2 points) Give one advantage and one disadvantage of synthetic fibers compared to natural fibers.
21. (1 point) Describe the differences between human, bat, and cow hair medulla.
22. (2 points) Hair development occurs in a(n) \_\_\_\_\_ order. In order, list the order of structure development.
23. (2 points) What is the role of the dermal papilla in the hair cycle?
24. (2 points) What effect does repeated bleaching have on hair structure, and which bond type is most affected?

## Chromatography/Spectroscopy

25. (10 points) Faker has discovered 6 suspicious smudges of blood (ink) near the site of the missing cannon minion. Help him determine which champion's ink matches the culprit! Use the provided samples to create 6 chromatography sheets. Staple these to the **FRONT** of your final exam. Label each sheet clearly with your team number in pencil.
26. (3 points) Calculate the R<sub>f</sub> of each chromatogram. Show work and label each chromatogram R<sub>f</sub>.
27. (2 points) Compare the chromatograms collected with our knowns. Who might these indicate?
28. (2 points) TLC, GC, and HPLC, are three forms of chromatography used for different purposes. Rank, in order from least to greatest resolution, these three chromatography methods.
29. (2 points) In TLC, how does the polarity of the solvent system affect the separation of a mixture of compounds such as benzene, toluene, and ethyl acetate? List the compounds in increasing order of polarity.
30. (1 point) Compare the polarity of acetone and dimethyl ether.

## Crime Scene Physical Evidence

31. (5 points) Faker compiled some fingerprints found at the nexus into a ten-print card. Label each of these fingerprints using their type and NCIC FPC Codes. The ten-print card can be found as Figure 1 in your image sheet. Some terms may or may not repeat.

RT:

RI:

RM:

RR:

RL:

LT:

LI:

LM:

LR:

LL:

32. (1 point) Write the primary classification number of the suspect.
33. (1 point) Write the NCIC FPC of the suspect.
34. (5 points) Why are fingerprint development techniques designed for porous surfaces not effective on non-porous surfaces? List two examples of each type of technique and what they react/adhere to.
35. (5 points) Faker has found some traces of DNA around the crime scene, but needs help identifying them! Using Figures 2 - 6, write down each DNA sequence. Each sequence may vary in length. Assume the right and left sides of the diagram correspond to 5' and 3', respectively.  
\*Key\* A:Green, T:Red, G:Black, C:Blue

36. (1 point) Faker and Keria discovered a trace of DNA near the missing cannon minion. Use Figure 7 to write out the DNA sequence shown on this PAGE gel. Be sure to label the 5' and 3' ends.
37. (2 points) What is an electropherogram and a consensus sequence? What is the difference between the two?
38. (3 points) What are the chemical components involved in the polymerization of a polyacrylamide gel, and how do they function?
39. (2 points) What is the difference between polyacrylamide and agarose gels when used for DNA electrophoresis?



## Analysis (TB)

40. (100 points) Who committed the crime? Include all relevant information. Cite motives and alibis for each suspect.